

Cognizant Digital Nurture 5.0

ReactJS Hands-On Exercise 6

Objective

Develop a React Router application named TrainersApp with trainer listing, navigation, and trainer details.

Theory

Need and Benefits of React Router

React Router enables client-side navigation without reloading pages.

Components in React Router

BrowserRouter, Routes, Route, Link, Navigate and useParams are common components.

Types of Router

BrowserRouter, HashRouter and MemoryRouter.

URL Parameters

useParams() retrieves route parameters such as trainer id.

Installation

npm install react-router-dom

Trainer.js

```
export default class Trainer{  
  constructor(  
    id,  
    name,  
    email,  
    phone,  
    technology,  
    skills){  
    Object.assign(this,{
```

```
id,  
name,  
email,  
phone,  
technology,  
skills});  
}  
}
```

TrainersMock.js

```
export const trainers=[  
  
  {  
  
    id:1,  
    name:'John',  
    email:'john@test.com',  
    phone:'9876543210',  
    technology:'React',  
    skills:'JS,React'},  
    {id:2,  
    name:'Mary',  
    email:'mary@test.com',  
    phone:'9876501234',  
    technology:'Angular',  
    skills:'TS,Angular'  
  }  
];
```

Home.js

```
export default function Home()  
  
{
```

```
return  
  
<h2>Welcome to Trainers App</h2>;  
  
}
```

TrainersList.js

```
import { Link } from "react-router-dom";  
  
import { trainers } from "../TrainersMock";  
  
export default function TrainersList() {  
  
  return (  
    <ul>  
      {trainers.map((trainer) => (  
        <li key={trainer.id}>  
          <Link to={` /trainer/${trainer.id}`}>  
            {trainer.name}  
          </Link>  
        </li>  
      ))}  
    </ul>  
  );  
  
}
```

TrainerDetails.js

```
import { useParams } from "react-router-dom";  
  
import { trainers } from "../TrainersMock";
```

```
export default function TrainerDetails() {

  const { id } = useParams();

  const trainer = trainers.find(
    (trainer) => trainer.id === Number(id)
  );

  return (
    <div>
      <h2>{trainer.name}</h2>

      <p>
        <strong>Email:</strong> {trainer.email}
      </p>

      <p>
        <strong>Phone:</strong> {trainer.phone}
      </p>

      <p>
        <strong>Technology:</strong> {trainer.technology}
      </p>

      <p>
        <strong>Skills:</strong> {trainer.skills}
      </p>
    </div>
  );
}
```

```
);  
  
}
```

App.js

```
import {  
  BrowserRouter,  
  Routes, Route,  
  Link, Navigate  
} from "react-router-dom";  
import Home from "./Home";  
import TrainersList from "./TrainersList";  
import TrainerDetails from "./TrainerDetails";  
export default function App()  
{  
  return (  
    <BrowserRouter>  
      <nav>  
        <Link to="/home">Home</Link>  
        {" | "  
        <Link to="/trainers">Trainers</Link>  
      </nav>  
      <Routes>  
        <Route  
          path="/"  
          element={<Navigate to="/home" />}  
        />  
        <Route  
          path="/home"  
          element={<Home />}  
        />  
      </Routes>  
    </BrowserRouter>  
  );  
}
```

```

/>

<Route
  path="/trainers"
  element={<TrainersList />}
/>

<Route
  path="/trainer/:id"
  element={<TrainerDetails />}
/>

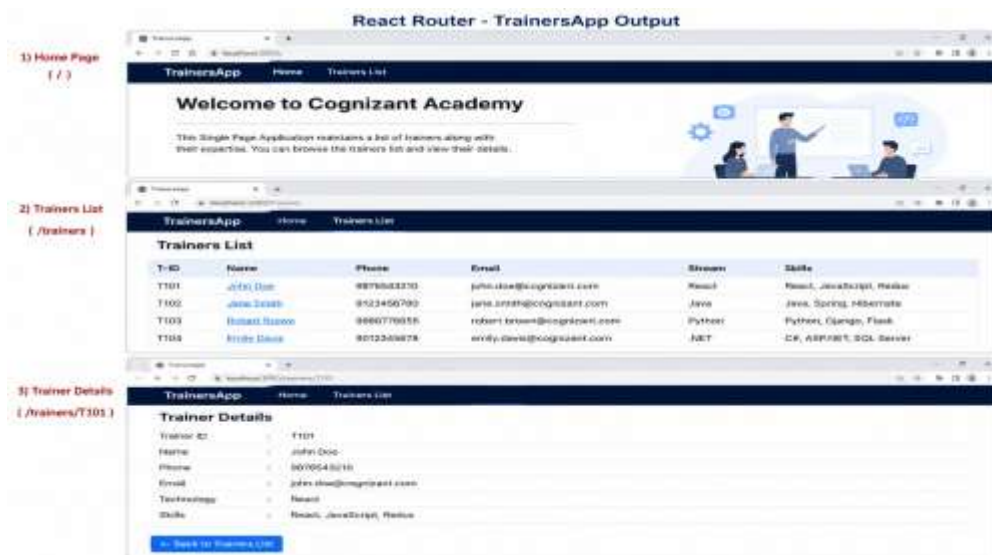
</Routes>

</BrowserRouter>

);
}

```

Output Screenshot:



Learning Outcome

Implemented React Router navigation and URL parameter passing.

Conclusion

Successfully created a SPA using React Router.